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Review

State-of-the-art-review on rice husk ash: A supplementary cementitious material in concrete

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ABSTRACT

Rice husk ash (RHA) is a good supplementary cementitious material for concrete production, as it requires low energy and emits negligible greenhouse gas during processing and service life as well as it shows high pozzolanic reactivity. The pozzolanic reactivity of RHA depends on its amorphous silica content, fineness, mix proportion, available alkaline media, and temperature because all of these factors are related to the dissolution of silica from RHA. Thus, incineration time, temperature, processing, and grinding of RHA should be controlled to acquire the desired level of pozzolanic reactivity. Depending on the filler effect and pozzolanic reactivity of RHA, concrete's strength varies. The RHA-blended cement concrete possesses dense microstructure, high mechanical performance, and enhanced durability against harsh environmental exposure. However, the cement replacement level by RHA should be within the optimum level, which is a factor of fineness and the mixed proportion of concrete. This review covered the current practice and guidelines of RHA-blended concrete. Simultaneously, this review also revealed a gap in in-depth investigations about the long-term durability and serviceability of reinforced RHA-blended concrete. Further research could lead to the application of RHA as a cost- and environmentally competitive alternative in the production of high-performance concrete. The summary and discussions provided in this paper will provide both direction and knowledge on the applications of greener and more sustainable RHA-blended concrete for researchers.

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1. Introduction

With the rising development globally as well as the ever-rising population, the demand for the construction of infrastructures and buildings is tremendously increasing. Most of the construction systems require natural raw materials, consume high energy, demand high cost, and produce waste during material processing and construction. Additionally, the negative environmental impacts of some construction material are increasing in some cases. Relatedly, the construction of buildings and infrastructures using cement-concrete is the most popular due to its low-cost requirement, outstanding durability, high mechanical strength and it is also the most convenient and easiest to apply (Nuaklong et al., 2020). However, cement production is a costly system; it requires limestone as raw materials, consumes high energy, and emits CO₂ during its production (Rattanachu et al., 2020). Approximately 5–8% of the global CO₂ emissions are generated by the production of ordinary Portland cement (OPC) (Khan et al., 2017; Sani et al., 2020). Increasing concern for environmental protection as well as energy conservation with minimal impact on the economy has motivated researchers to look for other alternatives of cement from waste and by-products. This leads to sustainable, green, and environmentally friendly construction.

The essential properties of supplementary cementitious materials (SCM) are having compatible bond-ability with aggregates (comparable to cement) and showing good pozzolanic behavior (Al-Kutti et al., 2019). Rice husk ash (RHA) has drawn significant attention toward the use of sustainable and environmental-friendly SCM in concrete (El-Sayed and El-Samni, 2006; Meddah et al., 2020). The RHA contains around 90% silica and shows excellent pozzolanic properties due to its high surface area, high amorphous nature, and compatibility with cement-concrete (Al-Kutti et al., 2019; Rodríguez de Sensale, 2010; Sandhu and Siddique, 2017; Sujivorakul et al., 2011).

The United States Department of Agriculture (USDA) estimated that the world rice production in 2019/2020 would be 499.31 million metric tons. In 2018/2019, this figure was 499.37 million metric tons (World agricultural production, 2020). During the milling of rice, each kg of rice would leave 0.28 kg of rice husk. Therefore, a huge amount of waste is being produced each year. These rice husks are used as fuel in several industries to produce heat energy, such as incineration and burning unit. After full combustion of rice husk, 20–25% RHA by weight is produced (Siddika et al., 2018). A very limited amount of the RHA is then used as a fertilizing agent in the field, and unfortunately, most of them are dumped in open landfills.

RHA contains non-crystalline silica and CaO, thus can be effectively used as SCM in concrete (Gomaa et al., 2017; Siddika et al., 2019). The utilization of RHA in concrete provides improved strength and durability properties, reduced materials cost due to cement savings, and environmental benefits related to the disposal of waste materials (Karthik et al., 2017). In recent research, RHA has been being used in the concrete mix as up to 100% replacement of OPC. The performance of concrete with RHA is dependent on the level of OPC replacement, the particle size of RHA, the chemical properties of RHA regardless of the aggregate, and the water/cement ratio of the concrete mix. However, for successful strength development, around 10–25% replacement of OPC is recommended (Sandhu and Siddique, 2017; Siddika et al., 2018).

This review included the discussion on all the parameters and activities of RHA when being used as SCM in concrete. Additionally, the present practice and future opportunities related to RHA-blended concrete are also analyzed based on current research. This review is a summary of in-depth information about RHA added to concrete, which will then serve as a guide for new researchers in this field.

2. Processing of rice husk ash

The physical properties and characteristics of RHA depend on its processing parameters, which include the burning method, separation process, and grinding. To produce RHA, open field burning or controlled incineration of rice husk can be applied. The amorphous silica and carbon contents of RHA are dependent on time and temperature of the incineration. Meanwhile, the pozzolanic reactivity of RHA is dependent on its amorphous content, specific surface area, and fineness of particles and those can be improved by adopting controlled combustion and grinding for application in structural concrete (Fapohunda et al., 2017). Generally, RHA, which contains approximately 85–95% of amorphous silica has shown compatible pozzolanic behavior in cement concrete.

Uncontrolled open field burning of rice husk results in high carbon content in RHA (Mboya et al., 2017; Nair et al., 2006), which can adversely affect the properties of concrete and produce highly crystalline microstructure. Complete burning at temperatures of 500 °C – 700 °C is recommended in previous studies (Ahmed and Adam, 2007; Alaneme et al., 2018), to have the highest amorphous silica with the maximum specific surface area of up to 150 m²/g of RHA particles. Additionally, after incineration, the separation of RHA is needed to ensure the elimination of any other impurities present, which could harm the pozzolanicity of RHA. Also, the

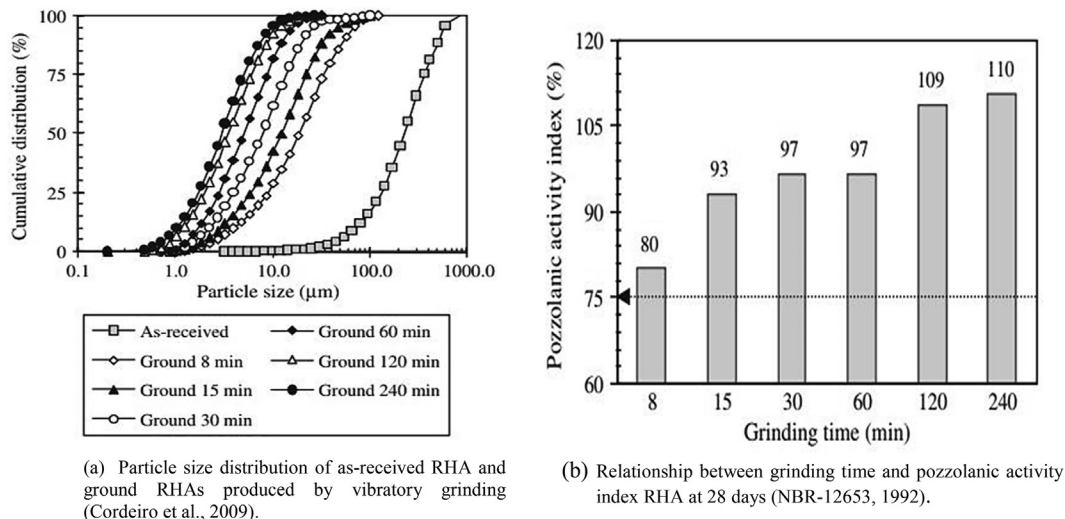


Fig. 1. Effect of grinding on RHA properties (Cordeiro et al., 2009).

Table 1

Physical properties of RHA.

References	Grinding time (minute)	Mean particle size (μm)	Specific gravity (gm/cm^3)	Fineness: passing 45 μm (%)	Specific surface area (m^2/g)
(Ganesan et al., 2008)		3.80	2.06	99	36.47
(Safuiddin et al., 2012)		6	2.1	–	2.33
(Habeb and Fayyadh, 2009)	90	63.8	–	–	–
	180	31.3	2.11	–	–
	270	18.3	–	–	–
	360	11.5	–	–	–

Table 2

Chemical compositions of RHA.

References	Chemical Constituents (weight, %)									
	SiO_2	Al_2O_3	Fe_2O_3	CaO	MgO	SO_3	Na_2O	K_2O	Others	Loss in Ignition
(Ganesan et al., 2008)	87.32	0.22	0.28	0.48	0.28	–	1.02	3.14	–	2.10
(Ramezaniapour et al., 2009)	89.61	0.04	0.22	0.91	0.42	–	0.07	1.58	–	5.91
(Cordeiro et al., 2009)	82.6	0.4	0.5	0.9	–	0.1	0.1	1.8	–	11.9
(Saraswathy and Song, 2007)	92.95	0.31	0.26	0.53	0.55	–	0.08	2.06	0.12	1.97
(Hossain and Anwar, 2014)	91.3	1.4	0.6	2.4	2.1	–	0.3	1.9	–	–
(Sua-iam and Makul, 2013)	93.44	0.21	0.18	0.76	0.43	0.16	0.05	1.98	–	1.27
(Chopra et al., 2015)	94.0	1.2	0.37	2.93	0.60	0.30	–	0.50	–	–
(Huang et al., 2017)	91.56	0.19	0.17	1.07	0.65	0.47	0.16	3.76	–	–
(Kannan and Ganesan, 2016)	87.89	0.19	0.28	0.73	0.47	–	0.66	3.43	–	4.36
(Xu et al., 2015)	91.71	0.36	0.90	0.86	0.31	–	0.12	1.67	–	3.13
(Le et al., 2014)	86.81	0.50	0.87	1.04	0.85	–	0.69	3.16	–	4.6
(Givi et al., 2010)	87.86	0.68	0.93	1.30	0.35	–	0.12	2.37	–	–
(Memon et al., 2011)	77.19	6.19	3.65	2.88	1.45	–	–	1.82	–	5.43
(Umasabor and Okovido, 2018)	87.22	0.70	1.68	2.12	1.18	0.04	0.20	1.12	0.46	1.06

metallic and alkali compounds (such as K^+) in the rice husk can be melted during incineration, which can reduce the specific surface area of RHA. Metallic ions can be separated from the rice husk using pre-treatment, such as acid-leaching (Adesina and Olutoge, 2019; Kwan and Wong, 2020). However, some recent research has demonstrated that RHA exhibits good performance even without any further processing (Siddika et al., 2018; Zerbino et al., 2011). Relatedly, mechanical grinding is important to achieve high specific surface area and fineness of the RHA particles.

The recommended minimum grinding time of RHA is 10–30 min by mechanical grinder (NBR-12653, 1992; Xu et al., 2015). However, the gradation and fineness of RHA for particular incineration and grinding conditions are different from others and possesses a specific impact on concrete properties. Decreasing the particle size is a common method to improve the pozzolanic activity of mineral

admixture. Van et al. (2013) reported that the particle size of RHA was decreased with increasing grinding time. However, grinding was associated with an approximately 12% loss in the ignition of RHA. As reported by Cordeiro et al. (2009), the grinding time of 120 min was sufficient to generate ultrafine ash with 6.8 μm average particle size, 33670 m^2/kg BET specific surface area, 109% pozzolanic activity index and 736 mg/g Chapelle activity.

Fig. 1a represents a typical relationship of grinding time and particle size distribution of RHA. It is observable that the particle size of RHA is decreasing with the increasing grinding time. Meanwhile, in its as-received condition, more than 80% of the RHA particles are coarser than 100 μm . Additionally, there is a close relationship between the specific surface area of RHA with grinding time, and consequently with its pozzolanic reactivity. This is evident in the research of (NBR-12653, 1992), and is represented in Fig. 1b.

3. Properties of rice husk ash

3.1. Physical properties

As-received RHA is a lightweight, bulky, and highly porous material with a density of around 180–200 kg/m³ (Zou and Yang, 2019). The specific gravity of RHA powder is between 2 and 2.5, depending on the fineness. As the pozzolanic reactivity of RHA depends on its fineness, the physical properties and gradation of RHA are important and they must be controlled to acquire the best performance. The typical physical properties of RHA are listed in Table 1.

3.2. Chemical composition

Typically, RHA contains more than 85% SiO₂, which is amorphous (Table 2). A small amount of CaO is also present in RHA, along with other oxides. However, the loss in the ignition of RHA products is high, due to the applied incineration, processing, and grinding. Most of the existing research has revealed that the amorphous nature of RHA is suitable to develop high strength in concrete (Antiohos et al., 2013; Djamaluddin et al., 2018). On the contrary, Bahrami et al. (2017) reported that crystalline RHA groups have shown higher mechanical strength than the amorphous one with a higher surface area. This can happen due to the crystal networks of the RHA particles, which provide high stiffness in the concrete matrix. However, this is a specific case. The RHA with amorphous silica and fine particles are most effective in showing pozzolanicity and binding action. Meanwhile, the other chemical contents of RHA have demonstrated a contributory effect on its binding property. The oxides of Fe, Al, and S have little effect on hydration, although Al improves the formation of aluminosilicate products in some cases, which also enhances concrete's strength. On the other hand, the alkali contents in RHA have illustrated a negative effect on both setting and hydration rate (Kashani et al., 2019).

3.3. Pozzolanic activity

The drop in electrical conductivity of a saturated solution of Ca (OH)₂ when a RHA sample is added can be taken as a measure of the pozzolanic activity of the samples (Ahmed and Adam, 2007). Initially, the conductivity of a saturated Ca(OH)₂ solution (200 ml, 40 °C) is measured. The 5 g of RHA sample is added to that solution and electrical conductivity is measured after two minutes of continuous stirring. The difference between the initial and final conductivities is calculated as a measure of pozzolanicity. This is identified as a rapid method to evaluate the pozzolanic activity of rice husk ash (Nair et al., 2006). According to ASTM C 618, any materials with 75% or better pozzolanic index can be used as SCM in concrete. As the pozzolanic reactivity of RHA is much

higher than the recommended value for grinding around 8 min, it has the highest potentiality to be used in concrete. However, regardless of the other mix design, water content, admixtures, and curing condition, the fineness of RHA still controls the pozzolanicity, thus grinding has an important impact. Furthermore, grinding methods also affect fineness, in which mechanical grinding is more efficient than the manual one. Calcination temperature and hence the degree of crystallinity were found to have a strong influence on pozzolanic activity. This could be attributed

to the fact that, at low temperatures, less crystalline silica is attained, which is more reactive than the highly crystalline silica obtained at higher temperatures. In this regards, Mboya et al. (2017) stated that the RHA from different sources achieves a pozzolanic activity index up to 88% at 7 days. Besides, as illustrated in Fig. 2, the pozzolanic activity of RHA could be enhanced with an increase in the grinding period, which results in the finer size of ash particles, and therefore, higher pozzolanic activity (NBR-12653, 1992).

3.4. Microstructure of RHA particles

The microstructure of RHA is dependent on the incineration, processing, and grinding methods. Typically, scan electron microscopic images (SEM) shows that RHA particles are generally irregular in shape, contain micro-porous surface and discrete widely. By incorporating 30 min of grinding, RHA particles began to show good dispersion, filler effect, and high pozzolanic reactivity (Xu et al., 2015). X-ray diffraction (XRD) analysis specifies that RHA comprises a significant amount of amorphous constituents with a very trivial quantity of crystalline parts such as cristobalite and sylvite, originated from the consumption of fertilizers by the rice husks (Sandhu and Siddique, 2017). A typical SEM image of RHA sample is shown in Fig. 2a-b. As observed from the figures, the as-received RHA particles are porous, non-uniform in dispersion, and coarse in nature. They are also not uniform in size and shape. However, upon grinding in a vibratory mill for about 120 min, the RHA particles became uniform in size, shape, and texture and dispersed uniformly with fewer pores trapped inside. Many other studies have also demonstrated a similar concept.

The XRD patterns of the as-received RHA samples illustrate that the RHA is partially crystalline and consists of cristobalite and amorphous compounds (Cordeiro et al., 2009). Typical patterns of XRD analysis of RHA are shown in Fig. 3a-b. The XRD of RHA without any treatment shows a hump shape that indicates the amorphous content. Also, the peaks of crystalline silica contents are visible at the Bragg 2-theta angle limit 25–27 (Chandrasekhar et al., 2003; Cordeiro et al., 2009; Khan et al., 2012). The grinding effect of RHA using XRD is represented by Fig. 3b. The results show that the grinding time of RHA has a very low impact on the XRD patterns. Still, any separation and loss in the constituents through

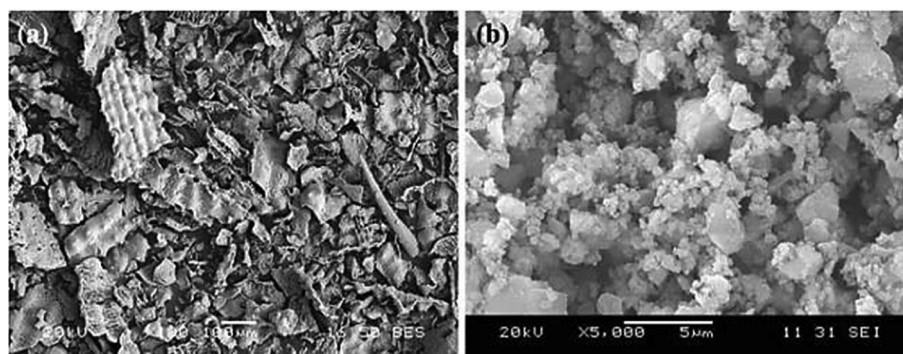


Fig. 2. SEM images of RHA (a) as-received; (b) ground for 120 min (Cordeiro et al., 2009).

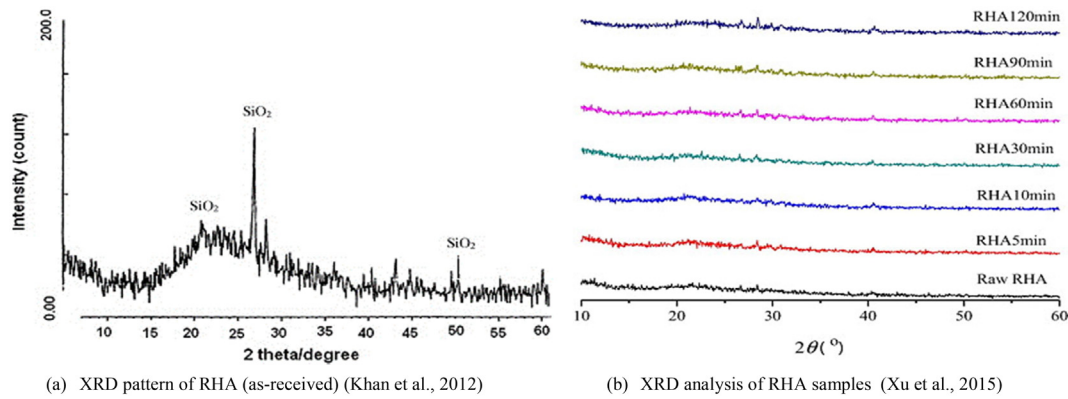


Fig. 3. Typical XRD pattern of RHA samples (Khan et al., 2012; Xu et al., 2015).

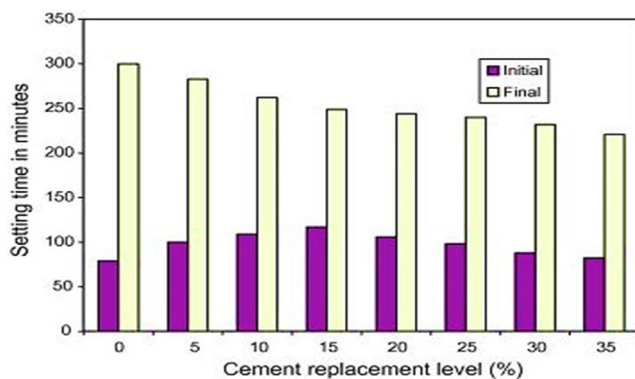


Fig. 4. Initial and final setting times of RHA-blended types of cement (Ganesan et al., 2008).

grinding could have a direct impact on the XRD patterns, which owes to the crystalline and amorphous nature. Besides, the most important is the amount of amorphous content in RHA, which is significant for the pozzolanic reactivity and to be a viable and effective SCM in concrete. In the as-received and ground states, a significant amount of amorphous contents present in RHA, while the partly crystalline phases are not so significant in reducing pozzolanicity.

4. Characteristics of cement blended with RHA

The activity of RHA starts when it is mixed with water and cement, and subsequently, alters the hydration and setting behavior. The setting time of RHA-blended cement increases with RHA content due to high dilution. As observed in Fig. 4 (Ganesan et al., 2008), the final setting time of OPC is 300 min, while 232 min are required for the OPC with 30% RHA. The linear reduction in final setting time could be caused by the accelerated hydration reaction and hardening of the paste due to the presence of RHA. The hydration of RHA-blended cement paste can be explained in the stages shown in Fig. 5. (Nguyen, 2011) explained that the first heat evolution peak of RHA-blended OPC and pure OPC is likely to be at the same time. However, the induction period is shorter in the RHA-blended binder due to the heterogeneous nucleation (Vieira et al., 2020).

The mechanisms and formation of intermediate calcium hydrates (CH) products are similar in both the RHA-blended cement and OPC and continue up to 3–8 h of hydration initiation. Resistance to CH formation is noticeable beyond that in the RHA-blended OPC. When the particles of RHA starts to participate by reacting with the CH products, heat evolution increases tremen-

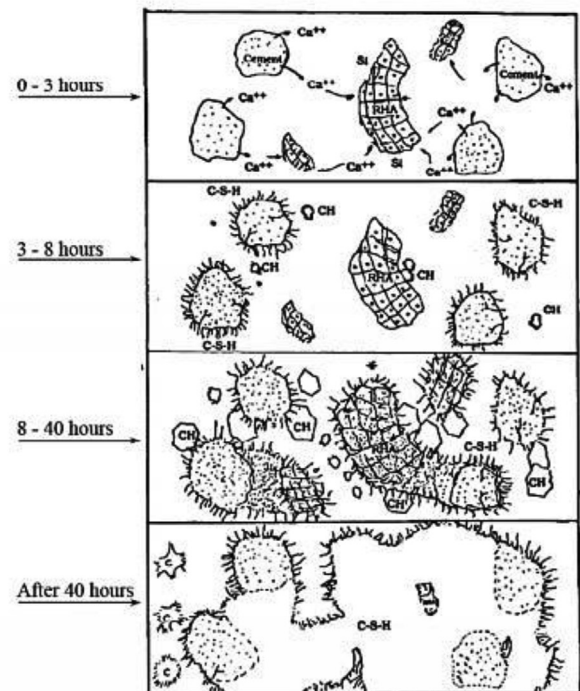
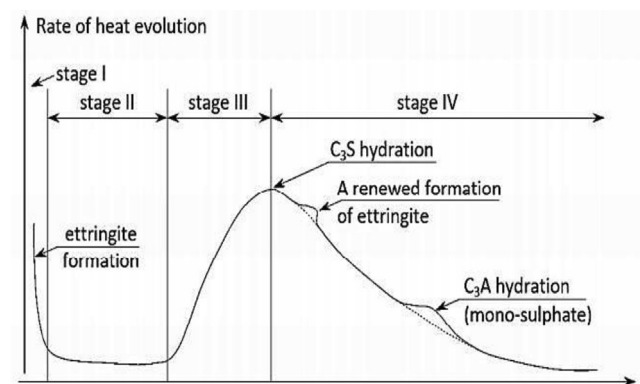


Fig. 5. Rate of hydration of cementitious materials (Nguyen, 2011).

dously (Van et al., 2014), owing to the formation of calcium silicate hydrates (C-S-H) products after the pozzolanic reaction of the silica from the RHA (Feng et al., 2004). The pozzolanic reaction of RHA continues over up to years, depending on the silica dissolution from the RHA and available CH (Mosaberpanah and Umar, 2019; Nair et al., 2006). The most suitable media to complete the perfect

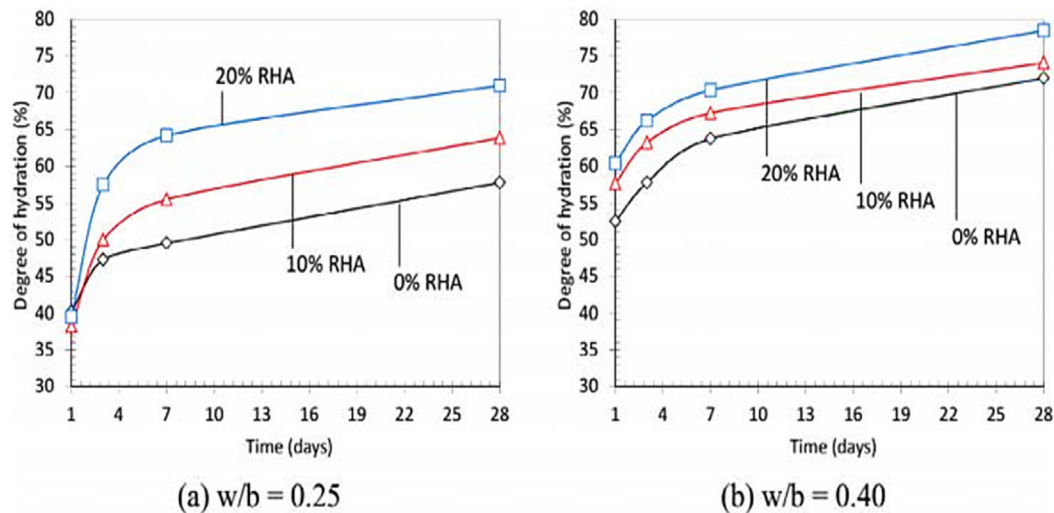


Fig. 6. Degree of hydration of cement in the RHA-modified samples (Nguyen, 2011).

hydration of RHA-blended cement is a high silica dissolution rate, alkaline media, a significant amount of CH products and a suitable temperature. In high alkaline media, silica dissolution is much more significant, and thus, is suitable for pozzolanic reaction. Early-stage pozzolanicity is not significant in the RHA sample because of the low amount of CH products formed in the cement paste and low alkalinity of the media. Once the optimum content of CH products and silica dissolution from the RHA occurs, the perfect hydration of RHA samples takes place. However, the main products of the hydration of the RHA-blended cement are C-S-H, ettringite, calcium tri-sulfo-aluminate hydrate, and their production is dependent on the available ions in the paste (Nguyen, 2011). The other important parameters are the particle size of RHA samples and the level of replacement with OPC. As the particle size of RHA decreases, the silica dissolution and pozzolanicity of it increases (Rodríguez de Sensale, 2006; Siddika et al., 2018). Therefore, a higher amount of C-S-H products are formed with a higher rate of hydration. Besides, the heat released increases during hydration. With the increasing amount of RHA, initially the pozzolanic reactivity increases; however, beyond the optimum level of cement replacement, the amount of CaO starts to decrease, and consequently, the formation of CH also decreases. Having said that, the optimum level of replacement of OPC is below 30% (Hwang and Chandra, 2008; Siddika et al., 2018; Zou and Yang, 2019).

It can be observed in Fig. 6 that hydration increased with RHA content, regardless of the water-cement ratio. Nonetheless, for the higher consistency level of the RHA-blended sample, a higher degree of hydration was observed at a higher water-

binder ratio. Typically, the consistency of cement increases with RHA content (Ganesan et al., 2008). The requirement of water also increases with the fineness and specific surface area of RHA and cement. RHA possesses porous structures, and thus, absorbs a higher amount of water, in comparison to the control paste (Nguyen, 2011). However, this water is released when the paste starts to harden. Thus, beyond the optimum level (30%) of cement replacement, the period of hydration will delay and increase the maximum secondary exothermal curve (Nguyen, 2011). Additionally, for excessively fine particles, when high dilution takes place, hydration can delay (Xu et al., 2015). The suitable mean particle sizes for hydration reactions are between 4.8 and 7.9 μm (Nguyen, 2011). Nonetheless, this hinges on the concrete mix characteristics. Besides, Vieira et al. (Vieira et al., 2020) concluded that RHA with 50% particle size 14–20 μm can effectively improve the

hydration behavior of the binder when replaced with 20% OPC and develop significant strength in concrete at 91 days. Given that, the coarse RHA particle is also effective with proper gradation and mix ratio in concrete.

5. Fresh properties of concrete with RHA

Workability is the measure of the ease and homogeneity of fresh concrete mix, with which it can be mixed, placed, consolidated, and finished (Fapohunda et al., 2017). Concrete's workability varies with the particle size, amount, water-cement ratio, characteristics of ingredients, and mix proportion. Regardless of other factors, the workability of concrete decreases with the increasing amount and fineness of RHA (Miller et al., 2019). RHA contains macro and mesopores inside its particles and with increasing fineness, its specific surface area also increases. Fine RHA subsequently absorbs a significant amount of water on its surface and stores the water in its pores, thus resulting in the decrease of free water and lowered slump value (Fapohunda et al., 2017). Moreover, the higher reactivity of RHA can be another contributing factor that lowers the flow of concrete (Le et al., 2015; Makul and Sua-iam, 2018; Memon et al., 2011). However, the self-consolidation of RHA particles also results in high resistance to segregation due to its filler effect, which increases the plastic viscosity and yields stress (Fapohunda et al., 2017). Furthermore, the excessive reduction in a slump and high plastic viscosity causes difficulties in molding and heterogeneous casting may results in, which is undesirable, and therefore, provides a cause of concern.

6. Physical properties of RHA-blended concrete

6.1. Density and microstructure

Owing to the high pozzolanicity, filler effect, and pore refinement of RHA particles, the density of concrete increases with the replacement of OPC by it. The increasing density of concrete is dependent on the fineness, amorphous silica content, and replacement level of RHA. As reported in the literature (Umasabor and Okovido, 2018), the hardened densities of concrete with 0%, 5%, 10% and 15% RHA are 2360–2400 kg/m^3 , 2360–2475 kg/m^3 , 2365–2515 kg/m^3 and 2050–2255 kg/m^3 , respectively. The reduction in the density of concrete with RHA beyond 15% replacement level could be attributed to the slow hydration and pozzolanic

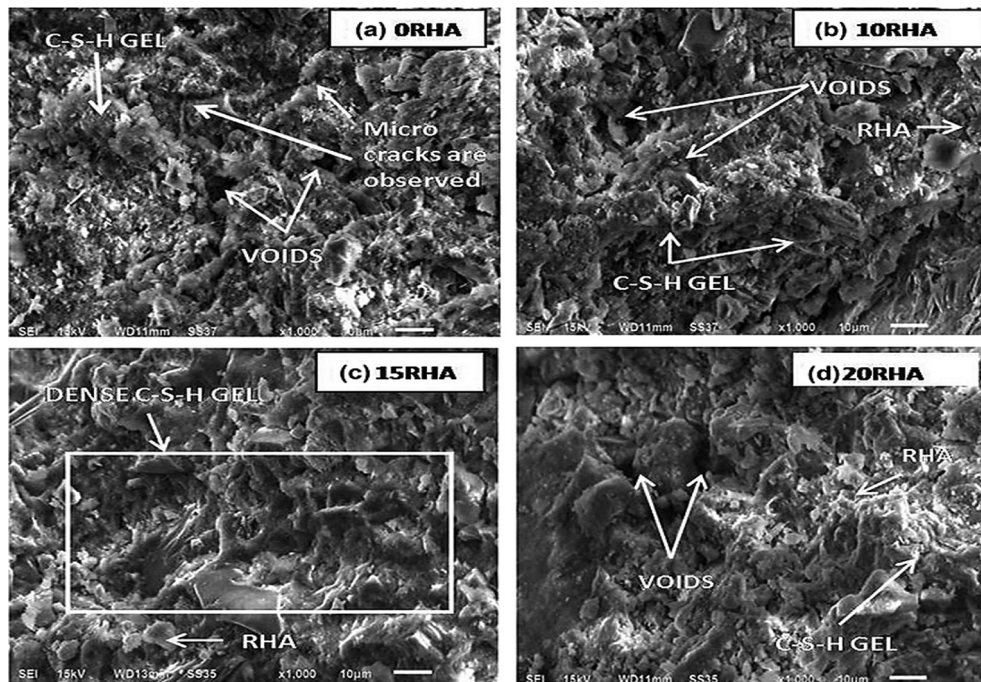


Fig. 7. SEM image of concrete at 28 days with 0–20% RHA (Chopra et al., 2015).

reaction of RHA-blended cement, which was described in Section 4. It is also important to note that the density of RHA-blended concrete can still be enhanced with up to 30% replacement level if proper silica dissolution and pozzolanic reaction take place.

The increasing amount of C-S-H products and decrease in the Portlandite content of the matrix can lead to the improvement of the pore structure of the concrete matrix, which is due to the high pozzolanicity and compatibility of RHA with cement (Chopra et al., 2015; Thomas, 2018). Additionally, the thickness of the interfacial transition zone (ITZ) around the coarser size aggregate in concrete with RHA is small, compared to the concrete without RHA. (Nguyen, 2011) also demonstrated that increased content and fineness of RHA made concrete pore structure finer, and hence, reduced the volume of large pores and also showed a reduction in the total porosity in the concrete. Meanwhile, Saraswathy and Song (2007) observed effective porosity values of 18.06%, 13.80%, and 11.89% for concrete containing 0%, 15%, and 30% RHA (as cement replacement), respectively. Indeed, the high reduction in porosity of the concrete with RHA has resulted in a densely packed microstructure (Chopra et al., 2015; Kartini et al., 2010). Fig. 7 shows the SEM image of the pore structure of concrete at 28 days with 0–20% cement replacement by RHA particles with a maximum size of <45 μm . The image clearly shows that after the hydration of RHA-blended cementitious material, a dense and uniform

microstructure of concrete can be formed. Increasing the content of RHA enhances the production of C-S-H gels and develops a denser concrete matrix, as illustrated in Fig. 7a–c. Beyond the optimum level of replacement, more amount and larger size of voids inside the concrete appeared. The density and micro-pores in the concrete also hinge on the other mixed ingredients and gradation of aggregates.

6.2. Shrinkage behavior

The shrinkage of concrete depends on its pores, density, and evaporable water inside the concrete. There is a contradiction in the concepts and interpretations of the shrinkage property of concrete. At the beginning of drying, the chemically unbound water starts to evaporate, which causes the concrete to shrink. As observed, the fine RHA particle shows good pozzolanic activity and this aid in the pore refinement in concrete. Based on the literature, concrete with SCM usually shows higher drying shrinkage and creep values (Habeeb and Fayyadh, 2009). However, the reason behind these concepts is unclear and unexplained. Nonetheless, the larger pores can entrap more temperature inside and reduce the heat transfer rate, and thus, reduces shrinkage, which would not happen in finer pores. On the contrary, due to the pozzolanic reactivity and dense microstructures of concrete with RHA as SCM, less shrinkage, and creeps have been observed (Chindaprasirt et al., 2004). Simultaneously, their long term behavior also demonstrates a negative effect. Besides, both shrinkage and creep also depend on air and water-filled voids in the concrete, aggregate proportion, and the characteristics of the ingredients. Up to the optimum level of cement replacement by RHA, the density of concrete increases, while voids decrease, with the fineness of RHA. Thus, it is expected to reduce both shrinkage and creep.

Ultra-high performance concrete experiences significant autogenous shrinkage (Shen et al., 2018), which is due to volume reduction during the self-hydration of cement. Additionally, dehydration occurs with a low water-cement ratio. The RHA particles have multiple layered microstructures (He et al., 2018; Shen et al., 2018), and thus, is capable of absorbing water. Besides, its micro surface

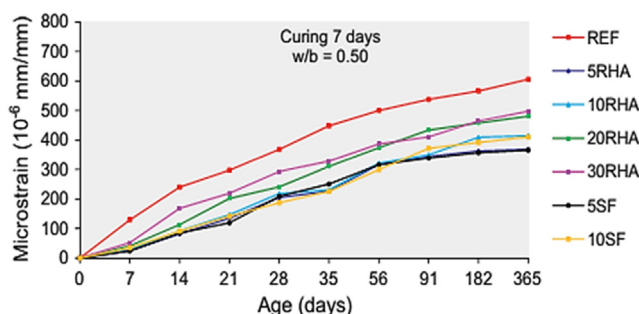


Fig. 8. Shrinkage strain of concrete at 7 days of curing (Gastaldini et al., 2014) (REF = 0% RHA, SF = silica fume).

compensates for the drop in the relative humidity of cement during hydration. Having said that, RHA acts as an internal curing agent that moderates high autogenous shrinkage of concrete (de Sensale et al., 2008; Mosaberpanah and Umar, 2019). A significant reduction in the shrinkage of concrete by using RHA was reported (Gastaldini et al., 2014), but a greater replacement level (>20%) lead to increased shrinkage (Fig. 8).

Additionally, for high curing temperatures, higher silica dissolution takes place, and a more pozzolanic reaction occurs, thus, a denser concrete matrix is formed. Furthermore, the high curing temperatures assist to reduce the chemically unbound water in the voids, which then reduces shrinkage. Ultra-high-performance concrete and self-compacting concrete with 20% RHA demonstrate remarkable lower shrinkage when cured at 65 and 90 °C (Van Tuan et al., 2011). As such, finer particles of RHA, high curing temperature, and optimum level of cement replacement can effectively reduce concrete shrinkage and creep, regardless of the other factors.

7. Mechanical properties of RHA-blended concrete

7.1. Compressive strength

Concrete compressive strength is proportional to its density (Wille et al., 2012; Yang et al., 2010). Additionally, it is dependent on the mix proportion, aggregate and cement properties, water-cement ratio, curing duration, and RHA replacement level. Regardless of the other factors, both the fineness and content of RHA are the major parameters that influence the strength development in concrete because these have influences on pozzolanic reactivity and hydration of binder. The recommended mean particle size of RHA to be used in high-performance concrete ranges from 3.6 to 9 μm , which is to be blended with Portland cement (Mosaberpanah and Umar, 2019; Nguyen, 2011). With the increasing fineness and content of RHA, there is a typical increase in the compressive strength of the concrete, and this is occurred by the physical filler effect, which is more than the pozzolanic reactivity at the early age of curing (<14 days) (Khan et al., 2012; Rodríguez de Sensale, 2006; Van Tuan et al., 2011). In the later stage of curing, the pozzolanicity increases significantly with the increasing amount of available CH products in the concrete. Pore size refinement also occurs in the concrete due to the pozzolanicity and filler effect of RHA. At the same time, the compressive strength increases as its pore sizes become smaller (Yin et al., 2006).

The optimum content of RHA varies depending on the RHA characteristics and types of binders. Typically, the optimum cement replacement level is considered to be around 20–30% for RHA (El-Dakroury and Gasser, 2008; Jamil et al., 2013; Muthadhi and Kothandaraman, 2013). Increments in the strength of approximately 25%, 33%, and 36% (for 7, 28, and 56 days respectively) were observed with the increments in RHA content (up to 15% cement replacement), as compared to the control mix. Moreover, concrete with 50–60% fine RHA (for cement replacement) showed comparable strength to the plain concrete. With increasing curing period, more hydration products are formed, and strength increased. However, when both the pozzolanic reaction and hydra-

tion completion reduced (due to any random causes), the strength development in the concrete also decreased (Chopra et al., 2015). The variation in strength depends on the curing age, and thus, it can be higher at a later age, i.e., 10% RHA at 3 days, 15% RHA at 7 days, and 10–20% RHA at 28 days (Le et al., 2014). Nonetheless, beyond the optimum level of replacement, concrete strength started to decrease with the amount of RHA. A typical variation in the compressive strength of concrete with RHA can be seen in Table 3. Note that the mix ratio and other parameters were different for each experiment. For example, using 5–10–14–18 graded aggregates instead of conventional aggregates, the strength in the RHA-blended concrete can be boosted by 4–14% (Rumman et al., 2020).

7.2. Splitting tensile strength

There is a very similar trend in the compressive strength and splitting tensile strength development of concrete with RHA. As observed in Fig. 9, the variation in the compressive strength of concrete with varying RHA contents is likely to be related to the variation pattern in the splitting tensile strength of that same concrete. Despite the other factors, by increasing both the contents and fineness of RHA, the splitting tensile strength of the concrete enhances due to the compactness and filling effect, and later, pozzolanic reaction (Chopra et al., 2015; Rodríguez de Sensale, 2006). As shown in Fig. 9, for the specific concrete mix, the optimum level of replacement varied because of the mix proportion, fineness, both physical and chemical characteristics of the ingredients, mixing, and curing condition. However, a typical range can be around 15–20%.

7.3. Flexural strength

The inclusion of fine RHA in concrete significantly improves its flexural strength (Rodríguez de Sensale, 2006). In the existing literature, it was reported that with fine RHA particles of up to 20%, the flexural strength of concrete improved by approximately 10–30% (Habeeb and Fayyadh, 2009; Naji et al., 2010). Owing to the high filler effect and uniform bond line development in the RHA-blended concrete, the improvement in post cracking behaviors also increased (Fapohunda et al., 2017). Additionally, Young's modulus of the 10–30% RHA-blended concrete was enhanced by 10–25%, which was due to the development of high crack bridging bonds and resistance to cracks up to a high deflection (Rodríguez de Sensale, 2006). A common representation for flexural strength and elastic modulus of RHA-blended concrete is shown in Fig. 10-a-b. However, all these improvements are dependent on the chemical characteristics and pozzolanicity of RHA, regardless of other factors. Depending on the incineration condition and burning temperature, both the carbon and amorphous silica in RHA are varied. Thus, to dissolve high amorphous silica and ensure good hydration of the RHA-blended cement, around 700 °C incineration temperature is effective, which is also mentioned for high flexural strength development (Bie et al., 2015). Additionally, silica dissolution from uniformly ground RHA is likely to be uniform and requires a similar environment to dissolve. Hence, a well-graded

Table 3
Compressive strength of concrete with varying RHA contents at different ages.

Age of Curing (Days)	Compressive Strength (N/mm ²) Cement replacement level by RHA (wt % of cement)							References
	0%	5%	10%	15%	20%	25%	30%	
7–200	9.4–22	8.5–40	7.2–27.2	1.2–24.2	–	–	–	(Umasabor and Okovido, 2018)
28	54	55.4	62.3	60.3	60.2	–	–	(Le et al., 2014)
7–28	27.22–36.45	31.32–36.49	30.45–37.43	31.52–37.38	31.64–37.71	33.09–39.55	33.53–37.80	(Saraswathy and Song, 2007)

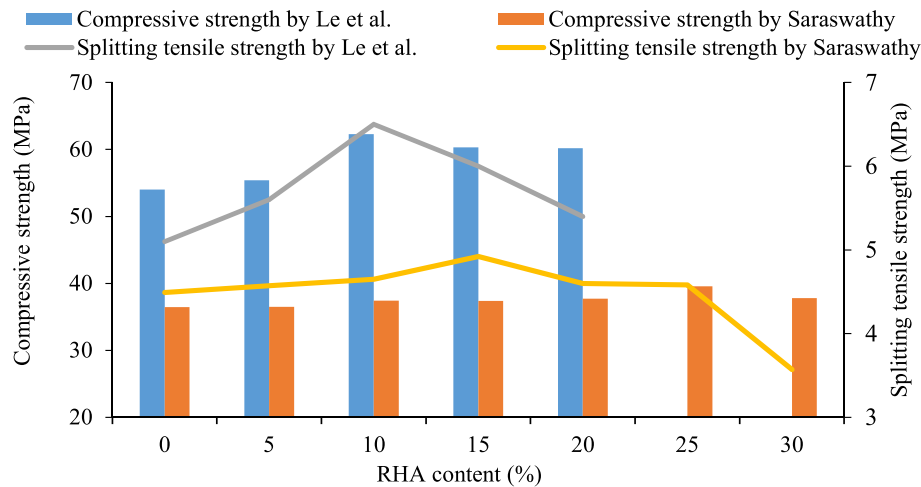


Fig. 9. Compressive strength-vs-splitting tensile strength of concrete with varying RHA contents at 28 days (Le et al., 2014; Saraswathy and Song, 2007).

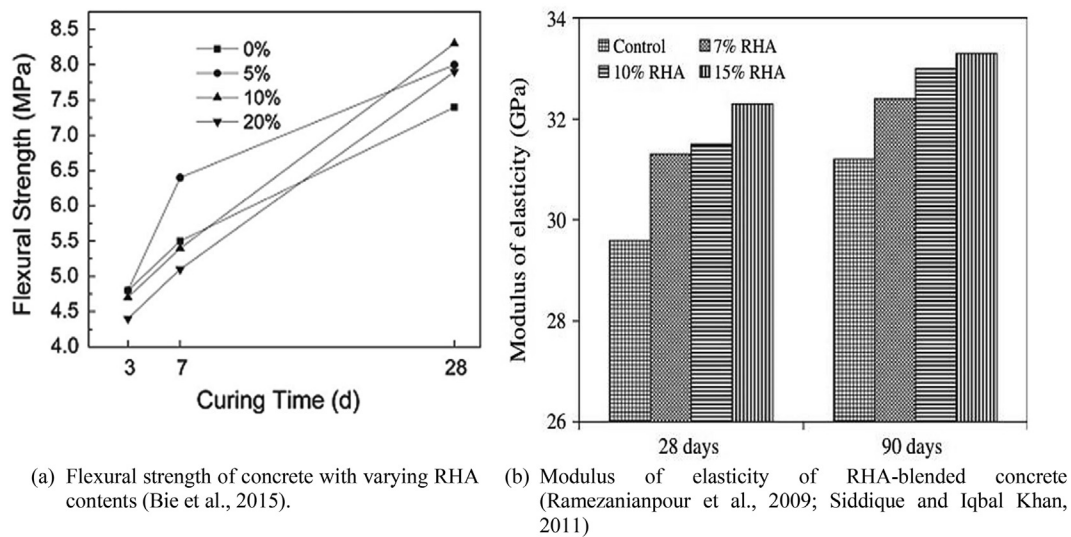


Fig. 10. Flexural performance of RHA-blended cement concrete (Bie et al., 2015; Ramezaniapour et al., 2009; Siddique and Iqbal Khan, 2011).

RHA can be effectively dissolve with the required silica for each curing stage. Given that, the grinding of RHA is another controlling parameter for strength development.

8. Durability properties

8.1. Permeability and absorption of water

The most common issue on durability arises from the penetration and absorption of liquids, ions, and gasses by the concrete due to the exposure environment. This causes the concrete in core structures to deteriorate as well as degrade both chemical and physical bonding. The penetration and absorption of liquid in concrete is proportional to its porosity. It has been demonstrated that concrete porosity can be reduced by adding up to 30% RHA (Saraswathy and Song, 2007), and this can also reduce both water penetration and absorption as well. Reduced water penetration and absorption have been observed in concrete with RHA in literature, as shown in Fig. 11 (Hamzeh et al., 2013; Ramezaniapour et al., 2009; Salas et al., 2009). However, the rate of water absorption is almost the same for 0–15% RHA-added concrete. As reported by Venkatanarayanan and Rangaraju (2015), compared to the con-

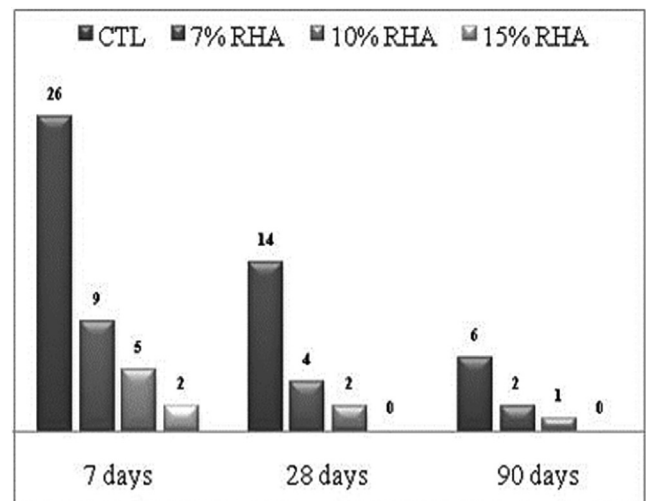


Fig. 11. Depth of water penetration (mm) at various ages of concrete without and with RHA (Ramezaniapour et al., 2009).

crete without RHA, a 13% and 12% decrease in water absorption occurred when the ground RHA was 7%, and 15%, respectively. However, the most significant factor is the fineness of RHA and its pozzolanic reactivity, which are important requirements to ensure the production of concrete with the lowest porosity. In the early age of concrete, penetration of water can assist in strength development in concrete by enhancing long term curing. But beyond the saturation limit and prolonged damped condition could reduce the strength, durability of concrete and can cause leaching and efflorescence of chemicals.

8.2. Chloride resistance

The penetration of any ions and solutions into the concrete is the initial step of corrosion initiation in steel reinforcement. The diffusivity of chloride ions in concrete depends on the porosity, pore connectivity within concrete (Al Mamun and Islam, 2017). Due to the dense microstructure of the RHA-blended cement concretes, its chloride ion diffusivity is lower than that of the OPC concrete (Salas et al., 2009). Additionally, RHA helps to modify and neutralize the effect of chlorine permeation and acid attack and corrosion of steel. According to the research of Al Mamun and Islam (2017) due to the fewer C-S-H gel dispersion around the outer surface of the casted concrete, the first 10 mm of the concrete cover provided a little barrier to chloride ion penetration and corrosion of reinforcement. Meanwhile, at 20–30 mm depth, the amount of penetrated ions was lower than that of the total threshold chloride ions content for reinforcement corrosion initiation. The core of concrete is generally denser and had a higher number of C-S-H products than that of the outer surface. Therefore, the production of more amount of C-S-H products in the RHA-blended binder is the main reason to acquire better chloride penetration and diffusion resistance (Chindaprasirt and Rukzon, 2008; Fapohunda et al., 2017). Accordingly, the inclusion of approximately 10–20% ground RHA can effectively reduce chloride penetration rate and corrosion in steel-reinforced concrete (Chindaprasirt and Rukzon, 2008; Rodríguez de Sensale, 2010). Consequently, concrete containing RHA may require less depth of cover to protect the reinforcing steel than that of the concrete containing OPC alone.

Moreover, regardless of the depth, the concentration of the penetrated chloride ions are also less in the RHA-blended cement concrete (in comparison to OPC concrete) (Siddique et al., 2016). As the age of the concrete increases, its hardened density increases as well, while the content of unreacted material decreases. As a

result, the penetration of chloride ion also decreases with time (Fig. 12). Moreover, depending on the gradation of aggregates and RHA, the pore refinement and reduction in porosity by RHA in concrete can be boosted (Rumman et al., 2020). Voids in the graded aggregate are compatibly filled up by the graded RHA, cement, and sand paste. Therefore, a dense microstructure and consequently high resistance to penetration of solutions will be achieved in concrete.

8.3. Carbonation resistance

Carbonation occurs due to the penetration of CO_2 into the concrete through its permeable pores. On the other hand, precipitation of carbonate takes place following the reaction with Ca(OH)_2 products. For RHA-blended cement concrete, the presence of unreacted Ca(OH)_2 is much less than the OPC concrete, which is due to the high pozzolanicity of the fine RHA and suitable curing condition of more than 28 days (Mosaberpanah and Umar, 2019; Nair et al., 2006). Additionally, up to the optimum level of replacement, the concrete with RHA shows high density and less porosity. Also, pore size refinement takes place as well, resulting in reduced pore connectivity. Consequently, CO_2 penetration also decreases. Therefore, for a replacement level of beyond 20%, the RHA-blended concrete shows better carbonation resistance than the normal concrete (Praveenkumar and Vijayalakshmi, 2019). However, by increasing the level of replacement and using coarser RHA particles, the carbonation depth in the RHA-blended concrete also increases during the early curing age (Sandhu and Siddique, 2017). In this regard, pore size, pore connectivity, and available CH products in the concrete are the major influencing parameters for increasing carbonation depth, and thus, this needs to be considered appropriate for RHA-blended concrete, as excessive carbonation can result in shrinkage and cause reinforcement corrosion. Relatedly, porous concrete is not recommended. It is also important to note that carbonation depth beyond the concrete cover will not affect the concrete strength and reinforcement corrosion initiation.

8.4. Sulfate attack

Sulfate attack can occur in two phases, either internally or externally. An internal attack is experienced when the sulfate ions from excess gypsum attach to the intermediate hydration products CH and subsequently, deteriorate the final products C-S-H gels. Internal sulfate attack and deterioration in concrete vary with the porosity and bond strength in hydration products (Tayeh et al., 2019). However, RHA improves the resistance of concrete to sulfate attack, which has been investigated by several researchers (Hossain and Anwar, 2014; Praveenkumar and Vijayalakshmi, 2019; Rodríguez de Sensale, 2010). Although the external sulfate attack and its chemistry are still unrevealed, it has been proven that the expansion due to sulfate attack of 15% RHA-blended concrete is much lower than that of the OPC concrete. Regardless, the expansion of concrete due to sulfate attack is dependent on the concentration of sulfate salts, types of salts, and other reaction media. Long term (1, 3, and 6 months) durability against a 5% MgSO_4 solution revealed a reduction in the compressive strength of concrete cubes, as a result of sulfate attack (Hossain and Anwar, 2014). Moreover, Rodríguez de Sensale (Rodríguez de Sensale, 2010), performed a sulfate attack test with a 5% sodium sulfate solution on RHA-blended concrete and found a similar conclusion. Additionally, the test results on RHA-blended cement paste revealed that at the early age (>15 days) of immersion of the paste into sodium sulfate solution (5%), more ettringite products were formed; thus, strength improvement occurs, but beyond this period, strength loss was also significant (Fig. 13). Further-

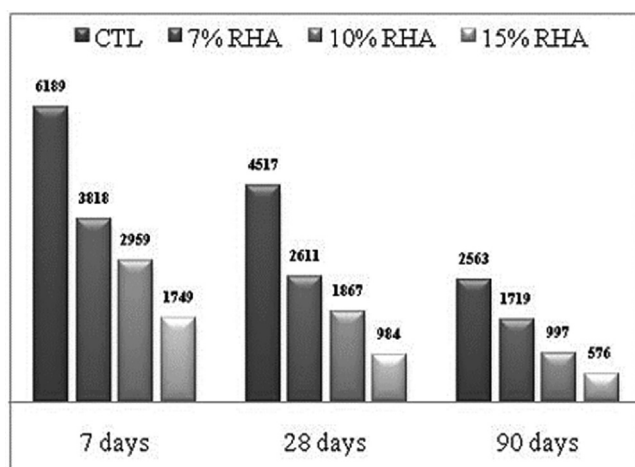


Fig. 12. Resistance to chloride ion penetration (coulomb) at various ages for control and RHA mixtures (Ramezaniapour et al., 2009).

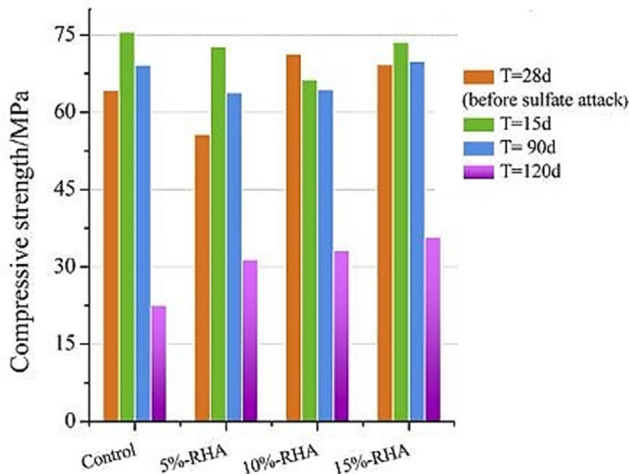


Fig. 13. Compressive strength of RHA-blended binder paste after sulfate attack (Hu et al., 2020).

more, (Chindaprasirt et al., 2007) concluded that RHA at a dosage of up to 40% cement replacement is very effective in providing sulfate resistance. The concrete with RHA possesses a lower pH level, and thus, reduces the harmful reactivity of sulfate ions up to a reliable limit. However, porous concrete is highly susceptible to sulfate attack, and thus, cement replacement should not be exceeded 20–30% to produce a dense concrete.

8.5. Alkali-silica reaction

The most vulnerable risk in concrete is the expansive gel formation, following the alkali-silica reaction (ASR). The alkali content in cement and aggregates reacts with the silica content, and RHA can negatively harm the durability of concrete by forming expansive gels. However, (Naji et al., 2010) reported that the incorporation of pozzolanic RHA by up to 15% cement is quite sufficient to control deleterious expansion due to alkali-silica reaction in the concrete, depending on the nature of the aggregates. The inclusion of RHA results in the entrapment of alkalis by the intermediate and supplementary hydrates. Also, it lowers the pH level in the concrete matrix. Consequently, the expansive gel formation by the alkali-silica reaction reduces because of the expansion of the mortar bar, which is sensitive to the pH level of the solution (Naji et al., 2010). The increasing fineness of RHA and curing temperature are also effective to enhance the resistance against expansion due to the alkali-silica reaction. It was proven that with fine RHA particles (<5 μm), only C-S-H products containing alkalis were formed and surrounded (Le and Ludwig, 2020).

Additionally, inner hydrates products with a high Ca/Si ratio also appear. However, for the coarser RHA particles (around 15 μm), two types of hydrates are formed. The first phase is where C-S-H containing alkalis forms a pozzolanic reaction rim around the RHA particles and these rims are like a semi-permeable membrane. Then, the penetration of Ca^{2+} , K^+ , Na^+ and OH^- ions results in the production of new ASR products. This is because being small-

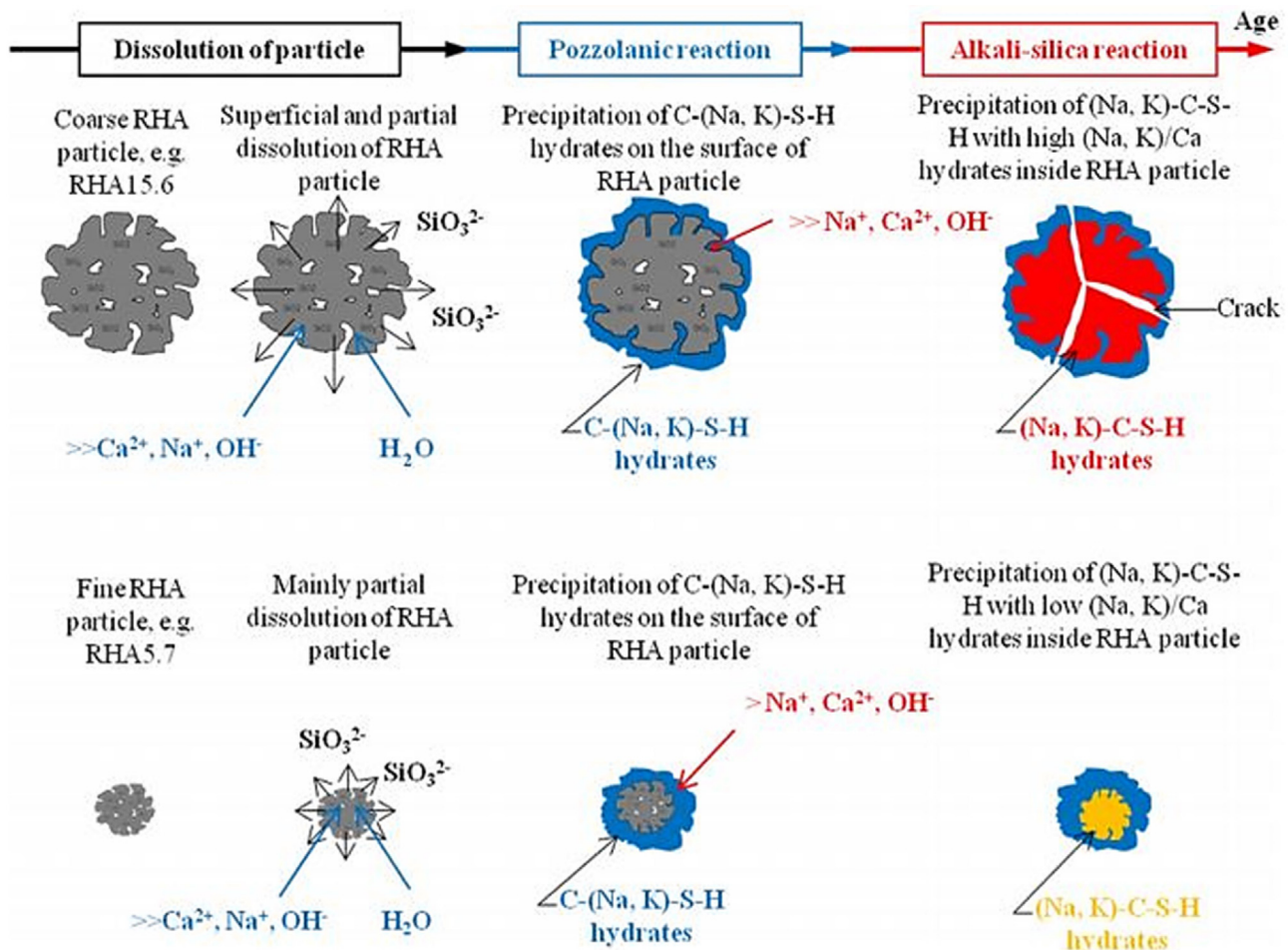


Fig. 14. Mechanism of successive reactions of RHA particles (Le and Ludwig, 2020).

ler in size, the take-up of Na^+ and K^+ ions are faster than the Ca^{2+} ions (Fig. 14). Therefore, ASR forms in the coarser RHA particles much quicker and in more volume, compared to that of the finer ones. Indeed, RHA particles of below $5\text{ }\mu\text{m}$ in size can effectively resist ASR expansion (Le and Ludwig, 2020). Ahmed et al. (Ahmed and Adam, 2007) found reliable durability in concrete with 12–15% RHA, which was cured for 14 days with NaOH solution at 80°C temperature. In another study, (Zerbino et al., 2011) found that the expansion of concrete containing ground RHA was within the limit of 0.04%. the ASR is inversely proportional to both pozzolanic reaction and hydration product formation. Given that, it is important to follow the guidelines for pozzolanic reactivity in regards to selecting fineness, replacement level, mixing, and curing condition of the RHA-blended concrete. Regardless, in porous concrete with high pH, the formation of expansive gels is unexpectedly beyond the allowable limit.

9. Limitations and further research

Given the desirable properties of RHA as an SCM, its application is not only limited to cementitious concrete but could also be extended in geopolymer concrete, pavement blocks, bricks, self-compacting concrete, high-performance nano-composites, and fiber reinforced concrete (Huang et al., 2017; Sandhu and Siddique, 2017; Siddika et al., 2020, 2019, 2018). Additionally, RHA could be effectively used to make an activator solution for alkali-activated binders. However, all these applications are under trial, and further deep investigations are required to increase the application widely. Moreover, in the current practice, the application of RHA-blended concrete in full-scale reinforced concrete structure has not been investigated under service and extreme loading conditions. Additionally, there are no established standard guidelines yet to prepare, process, and utilize RHA in large structures. In the existing literature, researchers have only used their thoughts to verify the optimum level of replacement of cement and fine aggregates by RHA.

Furthermore, the compatibility and long term durability of RHA-blended concrete has not been discussed in previous literature. The corrosion of steel reinforcement in RHA-blended concrete is needed to be investigated in water, chloride, sulfate, and acidic exposure for long term conditions. As the current literature only contains information on chloride ion penetration induced corrosion and its effect on concrete, but a detailed investigation on steel reinforcement is required. Besides, the information on the life cycle assessment of RHA-blended concrete is scattered, thus conclusions from those are required to be analyzed deeply. Moreover, both the processing and controlled incineration of RHA with around $700\text{--}800^\circ\text{C}$ temperature include additional cost when processing the material. Thus, alternative processing could be beneficial for the concrete industry as both the extraction and silica dissolution of RHA hinge on its amorphous nature and fineness. Alternative and supplementary additives like nano-silica, fibers can be added into RHA-blended cement to increase the strength of concrete (Alyousef et al., 2020; Sabapathy et al., 2019; Tawfik et al., 2020). For example, the alkaline solution compatible with RHA-blended cement could be incorporated as an admixture to increase the dissolution rate. However, without proper investigation, this cannot be justified. Additionally, with the high amount and coarser nature of RHA, a porous and less dense matrix of concrete can be formed. Nonetheless, the inclusion of nano-clay, short fibers, and nano-silica has shown the ability to enhance the density, shock, and tensile stress resistance capacity of concrete. Therefore, these additives with RHA-blended concrete could be another alternative sustainable material for future construction.

10. Conclusions

Utilization of RHA as SCM in the concrete industry offers sustainability, environmental-friendly, and energy-saving alternative of cement and enhances the use of recycled waste as construction materials. The production, processing, and usage of RHA should be followed by the guidelines and standard practices to achieve the best performance. Upon reviewing the existing literature on RHA-blended concrete, the following conclusions can be drawn:

- 1) RHA contains high amorphous reactive silica. Depending on the control over the incineration process, the chemical composition of RHA varies. A fully controlled incineration of rice husk around 700°C is suitable for the production of RHA with more than 80% amorphous silica. The fineness of RHA and its specific surface area are also important for the pozzolanic reactivity with cementitious materials. Thus, a proper grinding for sufficient time should be applied. The recommended minimum mechanical grinding time is approximately 30 min to achieve the desirable properties of ash; however, depending on the source of RHA, the grinding time can vary.
- 2) Finer particles of RHA release more silica and shows the better filler effect, thus, has higher pozzolanic reactivity. However, the particle size distribution of RHA can be selected based on its amorphous silica content.
- 3) The microstructure, density, and strength of concrete are directly related to the hydration and curing period of the concrete. At an early age, due to the slow pozzolanic activity of RHA, the rate of strength development is low. Subsequently, at the ages of 28 days and beyond, the high pozzolanic reactivity and filler effect significantly enhance the strength of concrete by providing a dense microstructure through the formation of additional C-S-H gels. The recommended cement replacement level is 30% by RHA particles with a mean size of $20\text{ }\mu\text{m}$ or less. However, beyond the optimum level of replacement, the reactivity in RHA would decrease due to the shortage of calcium hydroxide products, and consequently, less C-S-H products formed, which results in high pores in the concrete matrix and, therefore, lower strength.
- 4) The addition of RHA (up to 30% of cement replacement level) densify the concrete matrix and reduce the volume of voids, which results in a lower rate of water absorption and penetration of chemical ions into the concrete.
- 5) The application of RHA in concrete technology has promoted RHA as a building material, rather than just a complete waste. Consequently, RHA has demonstrated its ability to improve the strength and durability properties of concrete, in addition to the reduction in the cost of construction and carbon emission. Therefore, this chain can be regarded as a beneficial phase for the environment, construction industry, and economy. Having said that, investigations to extend the application of RHA waste to create a value-added material is welcomed and beneficial.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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